

An End-to-End AI Agent for Kaggle Competitions

Sinica: an end-to-end LLM agent benchmarked against two frozen reference agents on 20 Kaggle competitions

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1 · Motivation

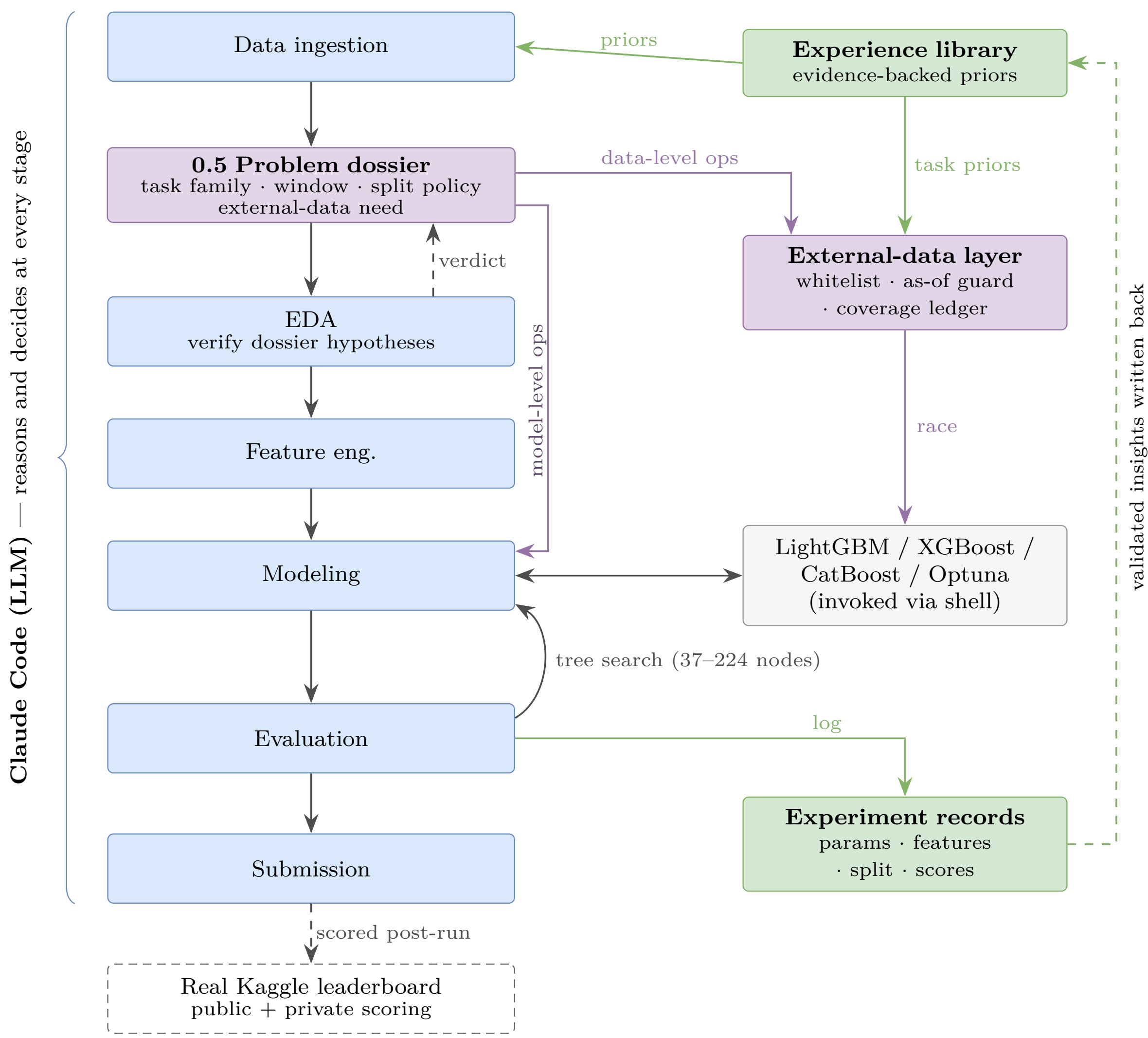
Challenges

- **Kaggle competitions** demand the full expert loop — validation design, feature engineering, model selection, ensembling — where the difference between top entries sits at the **4th–5th decimal** of the metric.
- With gaps that small, **local cross-validation is an unreliable compass**: a pipeline’s own validation score routinely disagrees with the real leaderboard, so no local number can be trusted to pick a winner.
- **Data visualization** (EDA, diagnostics, reporting) is essential but repetitive: every dataset needs distribution, drift, and correlation views before modeling can be trusted.

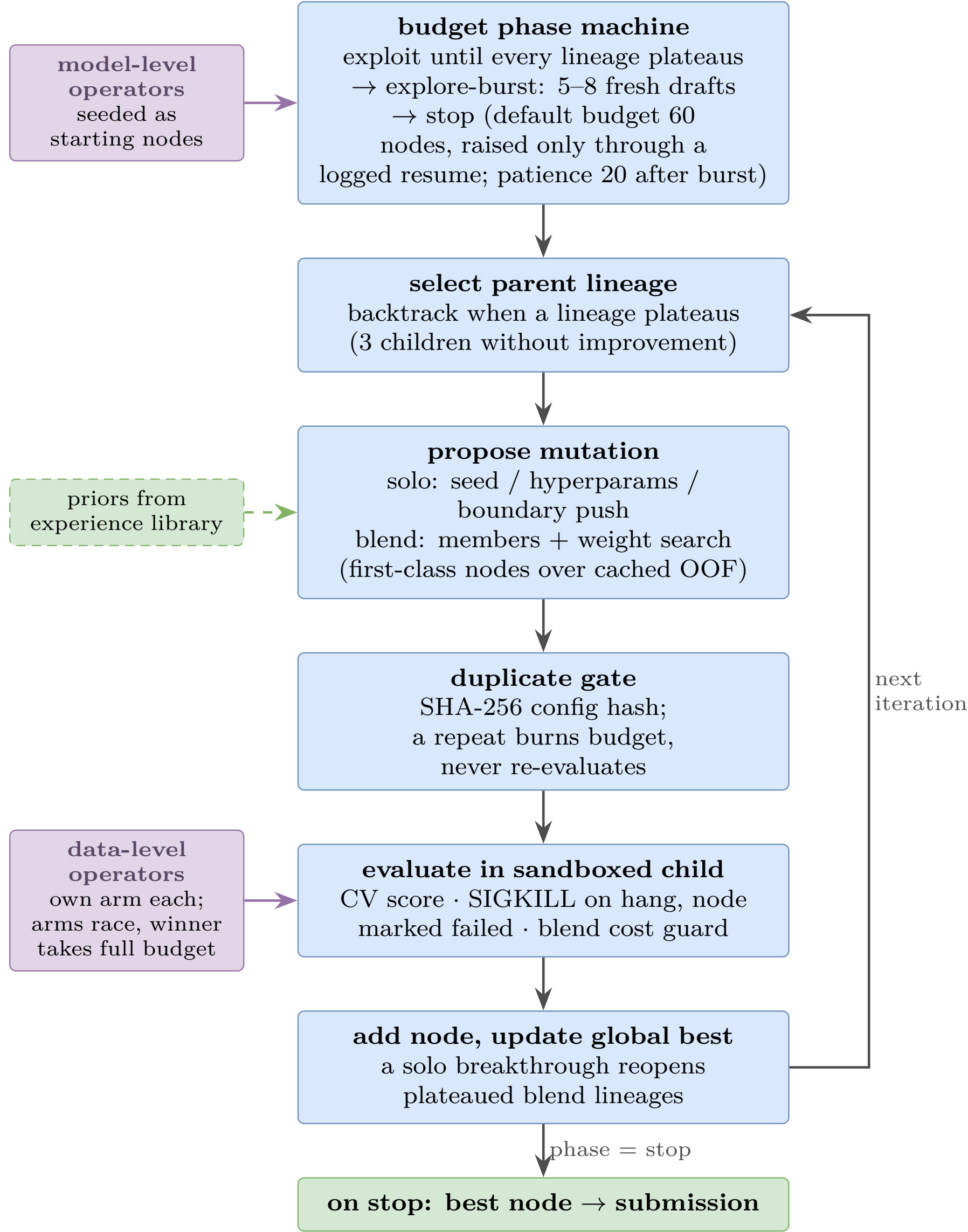
Why an AI agent

- Tireless, systematic search: 37–224 candidate configurations per competition, every experiment logged and reproducible.
- Runs the same disciplined protocol on 20 competitions serially — a human cannot hold this consistency.
- Records a per-dataset EDA pass and writes a per-competition report automatically, so every claim is grounded in an artifact.

2 · Workflow



The pipeline: the LLM reasons at every stage; the **problem dossier** classifies the task and injects external data *upstream*, before any modeling; submissions are scored post-run on the real leaderboard. Purple = injection layer · green = evidence loop · gray = Auto-ML tools; dashed = feedback or post-run.



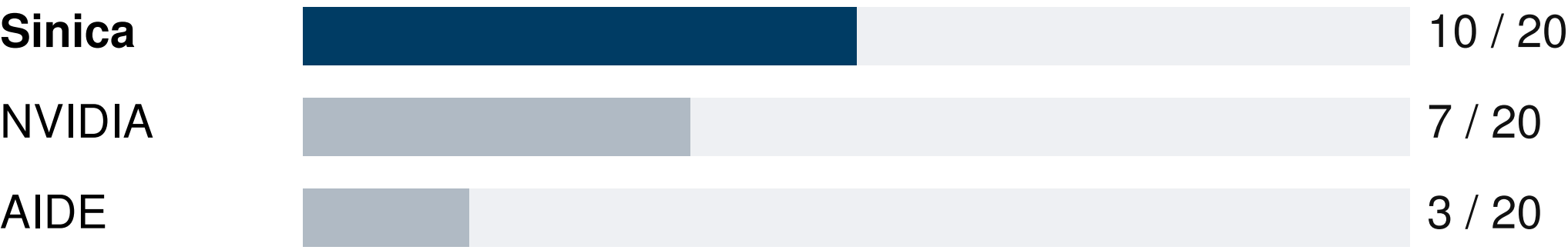
Tree-search loop: ensembles are first-class nodes; exploit until a lineage plateaus, then one forced exploration burst; the loop ends at the best-node submission.

3 · Results

Benchmark

All 20 competitions ran in sandboxes with a per-lane transcript audit, and all 20 are scored on the real leaderboard — as were the two reference agents. Each of the 60 head-to-head duels was judged against the score noise measured on Kaggle’s own public/private split: 38 decided, 22 too close to call. **Sinica leads the standing — 16 wins / 8 losses (66.7% of its decided duels)** against 42.3% for each reference agent.

Best private score



Private-leaderboard scores

Competition	Metric	Sinica	NVIDIA	AIDE
s3e1	RMSE ↓	0.55275	0.55550	0.55552
s3e3	AUC ↑	0.87157	0.89537	0.86863
s3e5	QWK ↑	0.57223	0.56974	0.58138
s3e7	AUC ↑	0.90441	0.90458	0.90109
s3e9	RMSE ↓	12.3319	12.2291	12.3067
s3e11	RMSLE ↓	0.29368	0.29624	0.29445
s3e14	MAE ↓	333.186	330.716	332.316
s3e16	MAE ↓	1.33822	1.34001	1.34224
s3e19	SMAPE ↓	49.4088	48.2656	48.3413
s4e11	Acc. ↑	0.94107	0.94076	0.94056
s5e1	MAPE ↓	0.08401	0.14594	0.17917
s5e10	RMSE ↓	0.05576	0.05588	0.05579
s6e1	Error ↓	8.71516	8.73239	8.72191
s6e2	AUC ↑	0.95505	0.95508	0.95516
afsis	MCRMSE ↓	0.48917	0.69855	0.49861
cat-in-the-dat	AUC ↑	0.80243	0.77084	0.80217
conway	MAE ↓	0.10414	0.11189	0.11065
tps-aug-2022	AUC ↑	0.58930	0.58730	0.59106
tps-jan-2022	SMAPE ↓	6.29895	4.63651	10.6664
tps-sep-2022	SMAPE ↓	28.4961	5.48351	9.72095

Bold = best of the three.

Three mid-complexity competitions

Competition	Metric	Sinica	NVIDIA	AIDE
us-patent (NLP)	Pearson r ↑	0.86658	0.86160	0.65821
ventilator (time series)	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC (imaging)	AUC ↑	0.93529	0.89822	0.88579

4 · Lessons learned

- **Put knowledge injection upstream.** At the blend stage it measured near-nothing and was abandoned; moved into problem identification it flags exactly the competitions that need external data, and **turns the canonical failure case around**.
- **No score until you submit.** While you iterate, local CV is all you see — unlike a task that can be graded on the spot — so an error in problem-level judgment leaves CV looking healthy while the leaderboard says otherwise.
- **Never mix agent versions in one table.** An architecture change invalidates every competition that ran before it: rebuild the run root, re-hash it, and start the benchmark over.
- **Isolate mechanically then audit transcripts.** Sandboxes stop leaks you anticipated; reading what the agent actually did catches the ones you didn’t — and it catches the auditor too: ours exempted every command starting with `uv`, so the one command it existed to catch was never examined.

5 · Weaknesses of AI-heavy research

- **No engineering judgment at design time.** The code is fast and accurate, but nothing in the loop can tell whether an architecture will hold until it runs — and by then a weak structure has started accumulating debt that grows with every competition added to it.
- **The question gets lost in the detail.** Heavy AI use pulls a write-up toward the technical detail it can report exhaustively, and away from the one thing the reader came for.
- **AI writing habits survive review.** Redundant heading suffixes and invented terminology come back in every draft; keeping them out is manual work, not a setting.